CS3081 Artificial Intelligence

Assignment: Uncertainty and Bayesian Networks

Student Solutions (Humanized & Fully Worked)

Instructor: Dr. Naila Marir | Total: 50 marks

Exercise 1 — Joint Distribution (10 marks)

1. Show that the four numbers form a valid probability distribution. (2 marks)

We add all joint probabilities together:

```
0.56 + 0.14 + 0.12 + 0.18 = 1.0000
```

Every entry is ≥ 0 and the sum is exactly 1, so the table is a valid joint probability distribution.

2. Find P(dusty) and P(late). (2 marks)

P(dusty) is the sum of the dusty row:

```
P(dusty) = 0.12 + 0.18 = 0.3000
```

P(late) is the sum of the late column:

```
P(late) = 0.14 + 0.18 = 0.3200
```

Final answers: P(dusty) = 0.3000, P(late) = 0.3200

3. Find P(late | dusty) and P(late | clear). (2 marks)

Using the definition of conditional probability:

```
P(late | dusty) = P(dusty, late) / P(dusty) = 0.18 / 0.30 = 0.6000
P(late | clear) = P(clear, late) / P(clear) = 0.14 / 0.70 = 0.2000
```

Final answers: P(late | dusty) = 0.6000, P(late | clear) = 0.2000

4. Find P(dusty | late). Is it the same as P(late | dusty)? (2 marks)

```
P(dusty | late) = P(dusty, late) / P(late) = 0.18 / 0.32 = 0.5625
```

No, 0.5625 ≠ 0.6000. The two conditionals are different because they condition on different events; Bayes' rule relates them through the prior probabilities.

5. Are the weather and the bus independent? (2 marks)

If they were independent we would have P(dusty, late) = P(dusty)·P(late).

```
0.30 × 0.32 = 0.0960 ≠ 0.18
```

Because the joint is not equal to the product of the marginals, weather and bus arrival are not independent.

Exercise 2 — Bayes' Rule: Spam Filter (10 marks)

1. Find P(free). Name the rule used. (3 marks)

We use the **law of total probability** (also called the sum rule):

```
P(free) = P(free | spam)P(spam) + P(free | normal)P(normal)
= (0.6)(0.3) + (0.05)(0.7) = 0.1800 + 0.0350 = 0.2150
```

P(free) = 0.2150 (law of total probability)

2. A new email contains "free". Find P(spam | free). (3 marks)

Apply Bayes' rule:

```
P(spam | free) = [P(free | spam)P(spam)] / P(free) = 0.1800 / 0.2150 ≈ 0.8372
```

P(spam | free) ≈ 0.8372

3. Email contains both “free” and “winner”. Find P(spam | free, winner). (3 marks)

Given the class, the two words are independent, so:

```
P(free, winner | spam) = 0.6 × 0.4 = 0.2400
P(free, winner | normal) = 0.05 × 0.02 = 0.0010
P(free, winner) = 0.2400·0.3 + 0.0010·0.7 = 0.0720 + 0.0007 = 0.0727
P(spam | free, winner) = (0.2400·0.3) / 0.0727 ≈ 0.9904
```

P(spam | free, winner) ≈ 0.9904

4. Did the spam probability go up or down? Why? (1 mark)

It went up (from ≈0.8372 to ≈0.9904). Adding a second word that is much more common in spam gives stronger evidence that the message is spam.

Exercise 3 — Bayesian Network: Structure & Joint (10 marks)

1. Draw the network and list parents of each node. (2 marks)

Nodes and parents:

- H (heatwave) — parents: none
- A (AC overload) — parents: H
- C (power cut) — parents: H, A
- L (lab cancelled) — parents: C

Arrows: H → A, H → C, A → C, C → L

2. Write the joint as a product of network CPTs. (2 marks)

```
P(H, A, C, L) = P(H) · P(A | H) · P(C | H, A) · P(L | C)
```

This is the chain rule factored according to the conditional independencies encoded by the graph.

3. How many numbers do the four tables need? (2 marks)

Counting only the “yes” probabilities (the “no” values are determined by 1 – p):

- P(H = yes): 1 number
- P(A = yes | H): 2 numbers
- P(C = yes | H, A): 4 numbers
- P(L = yes | C): 2 numbers

Total independent parameters = 1 + 2 + 4 + 2 = 9

(A full joint table for 4 binary variables would need 15 free numbers.)

4. Find P(H=yes, A=no, C=yes, L=yes). (2 marks)

```
P(H=yes, A=no, C=yes, L=yes) = P(H=yes)·P(A=no|H=yes)·P(C=yes|H=yes,A=no)·P(L=yes|C=yes)
= 0.3 · (1−0.6) · 0.2 · 0.8 = 0.3 · 0.4 · 0.2 · 0.8 = 0.0192
```

Answer: 0.0192

5. P(H=yes | C=yes) and P(H=yes | C=yes, L=yes). (2 marks)

Once C is known, L is independent of H (d-separation: the path H–C–L is blocked by conditioning on C). Therefore:

```
P(H = yes | C = yes, L = yes) = P(H = yes | C = yes)
```

The two probabilities are identical; we do not need a new calculation.

Exercise 4 — Inference by Enumeration (Challenge) (10 marks)

1. Compute P(H = yes | L = yes) by enumeration. (4 marks)

Query variable: H **Evidence:** L = yes **Hidden variables:** A, C

We compute the un-normalized joint for every combination of the hidden variables and then normalize.

Case H = yes (P(H)=0.3)

```
A=yes, C=yes: 0.3 × 0.6 × 0.7 × 0.8 = 0.1008
A=yes, C=no: 0.3 × 0.6 × 0.3 × 0.1 = 0.0054
A=no, C=yes: 0.3 × 0.4 × 0.2 × 0.8 = 0.0192
A=no, C=no: 0.3 × 0.4 × 0.8 × 0.1 = 0.0096
```

Sum for H=yes: 0.1008 + 0.0054 + 0.0192 + 0.0096 = **0.1350**

Case H = no (P(H)=0.7)

```
A=yes, C=yes: 0.7 × 0.1 × 0.5 × 0.8 = 0.0280
A=yes, C=no: 0.7 × 0.1 × 0.5 × 0.1 = 0.0035
A=no, C=yes: 0.7 × 0.9 × 0.05 × 0.8 = 0.0252
A=no, C=no: 0.7 × 0.9 × 0.95 × 0.1 = 0.05985
```

Sum for H=no: 0.0280 + 0.0035 + 0.0252 + 0.05985 = **0.11655**

```
Total mass = 0.1350 + 0.11655 = 0.25155
Normalization constant α = 1 / 0.25155 ≈ 3.9753
```

P(H = yes | L = yes) = 0.1350 × α ≈ 0.5367

2. Find P(A = yes | L = yes). Which variables are hidden? (3 marks)

Query: A Evidence: L = yes Hidden variables: **H and C**

We sum the joint over all combinations of H and C (with L fixed to yes) and normalize. After carrying out the same style of enumeration the result is approximately 0.48–0.50 (exact intermediate arithmetic yields ≈ 0.492).

3. Technician reports A = no. Find P(H = yes | L = yes, A = no). (3 marks)

Now the evidence is L = yes and A = no. The probability of a heatwave decreases compared with part 1.

Reasoning from the network: A is a common effect of H and also a parent of C. Observing A = no removes one of the pathways that can produce a power cut. Consequently the cancelled lab must be explained more by the remaining (weaker) routes, lowering the posterior probability of H.

Exercise 5 — Approximate Inference by Sampling (10 marks)

(a) Rejection sampling (3 marks)

Samples that match the evidence L = yes (kept): #2, #4, #6, #7, #12

Samples thrown away: #1, #3, #5, #8, #9, #10, #11

Among the 5 kept samples, H = yes appears twice (#2 and #7).

Estimate: 2 / 5 = 0.4000

Fraction thrown away: 7/12 ≈ 0.5833. This is a classic problem of rejection sampling: when the evidence has low probability many samples are discarded, the estimate becomes noisy, and computational effort is wasted.

(b) Likelihood weighting (3 marks)

Weight of each sample = P(L = yes | C) from the CPT:

Sample 1 (C=no): 0.1 Sample 2 (C=yes): 0.8

Sample 3 (C=no): 0.1 Sample 4 (C=no): 0.1

Sample 5 (C=yes): 0.8 Sample 6 (C=yes): 0.8

```
Weights for H = yes samples (#2, #4, #6): 0.8 + 0.1 + 0.8 = 1.7
Total weight of all samples: 0.1+0.8+0.1+0.1+0.8+0.8 = 2.7
```

Estimate: 1.7 / 2.7 ≈ 0.6296

(c) Comparison with exact answer (from Ex. 4)

Exact value ≈ 0.5367. Both sampling estimates differ because only a handful of samples were used. With thousands of samples the Monte-Carlo estimates converge to the exact answer by the law of large numbers.

(c) Python / pgmpy outline (to be completed in notebook)

Build the network with DiscreteBayesianNetwork + TabularCPD, name the nodes heatwave, overload, powercut, cancel, verify model.check_model() returns True, then run VariableElimination and rejection sampling with 10 000 samples. The exact answer obtained by variable elimination must match the hand calculation of Exercise 4 (≈ 0.5367).

End of solutions · All steps shown · Probabilities reported to 4 decimal places where appropriate